UNIT 1 Kinematics

1.1 Scalars and Vectors in One Dimension
1.2 Displacement, Velocity, and Acceleration
1.3 Representing Motion
1.4 Reference Frames and Relative Motion
1.5 Vectors and Motion in Two Dimensions

1.1 Scalars and Vectors in One Dimension

Scalar: a physical quantity that is completely described by its magnitude. Examples of scalars are volume, density, speed, energy, mass, and time.

Vector: in physics, a quantity that has both magnitude and direction. It is typically represented by an arrow whose direction is the same as that of the quantity and whose length is proportional to the quantity's magnitude.

Displacement: the change in position in some time interval.

(Displacement is a VECTOR QUANTITY)

Distance: the length of a path followed by a particle.

(Distance is a SCALAR QUANTITY)

Average Velocity : a particle's displacement divided by the time interval during which that displacement occurs

$$\bar{v} \equiv \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

Average Speed : the total distance traveled divided by the total time interval required to travel that distance

$$\text{Avg speed} = \frac{\text{total distance}}{\text{total time}}$$

Avg Velocity is a VECTOR QUANTITY

Avg Speed is a SCALAR QUANTITY

1.2 Displacement, Velocity, and Acceleration

Position: Location of a particle with respect to a chosen reference point.

Displacement: The change in position in some time interval.

$$\Delta x = x_f - x_i$$

Average Velocity: A particle's displacement divided by the time interval during which that displacement occurs.

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

$$\Delta x = \bar{v} \cdot t$$

1.2 Displacement, Velocity and Acceleration

Example 1:

Position	t(s)	x(m)
A	0	30
B	10	52
C	20	38
D	30	0
E	40	-37
F	50	-53

Find the displacement, average velocity of the object between positions A and F.

1.2 Displacement, Velocity, and Acceleration

Average Acceleration: The change in velocity divided by the time interval during which that change occurs

$$\vec{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}$$

Example 2

The velocity of a particle moving along the x-axis varies in time according to the expression $v_x = (40 - 5t^2)\ m/s$,

Where t is in seconds.

Find the average acceleration in the time interval t = 0 to t = 2.0s.

1.3 Representing Motion

Uniform Acceleration Motion (UAM):

A motion in a straight line with a constant acceleration and zero difference.

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}$$

$$\Delta v = at \text{ (for constant a)}$$

$$v = v_i + at \text{ (for constant a)}$$

$$\bar{v} = \frac{v_i + v_f}{2} \text{ (for constant a)}$$

1.3 Representing Motion

Uniform Acceleration Motion (UAM):

A motion in a straight line with a constant acceleration and zero difference.

$$\Delta x = \bar{v}t = \left(\frac{v_i + v_f}{2}\right) t$$

$$v = v_i + at$$

$$\boxed{\begin{aligned} \Delta x &= v_i t + \frac{1}{2} a t^2 \\ \Delta x &= v_f t - \frac{1}{2} a t^2 \\ v^2 &= v_i^2 + 2a\Delta x \end{aligned}}$$

1.3 Representing Motion

Uniform Acceleration Motion (UAM):

A motion in a straight line with a constant acceleration and zero difference.

Example 3

A jet lands on an aircraft carrier at 140 mph (63 m/s).

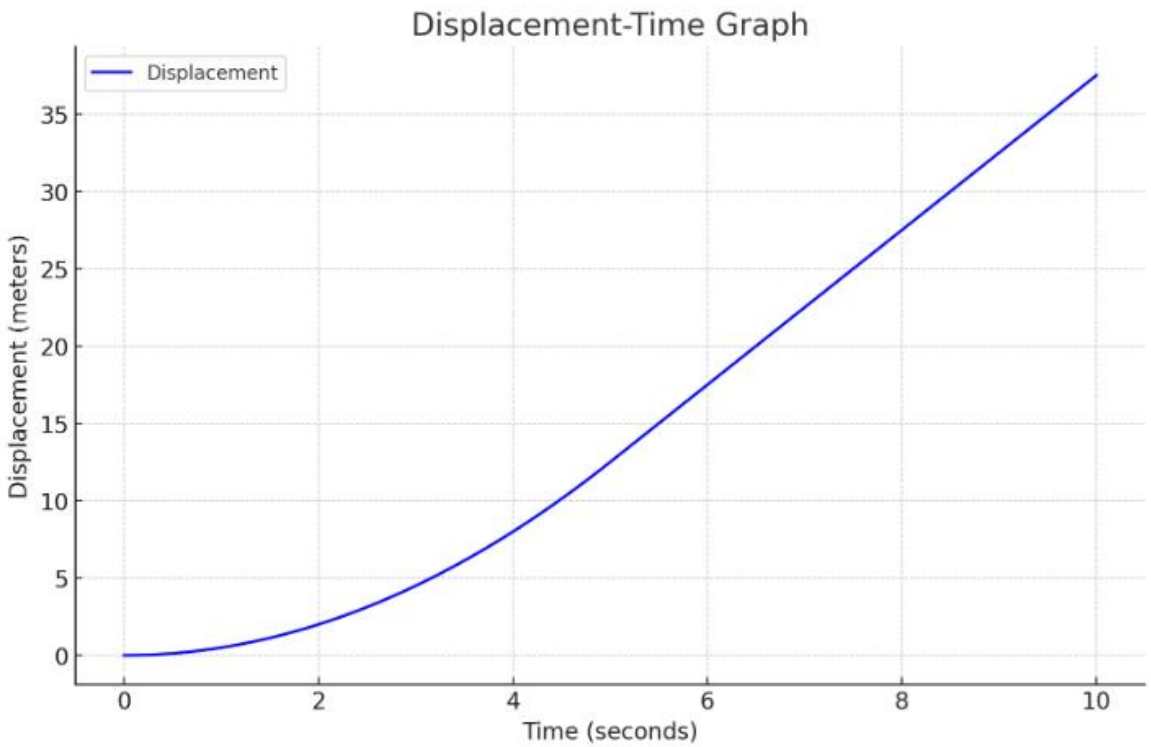
(a) What is its acceleration (assumed constant) if it stops in 2.0 s due to an arresting cable that snags the airplane and brings it to stop?

(b) If the plane touches down at position $x_i = 0$, what is the final position of the plane?

1.3 Representing Motion

The Graph:

1) Displacement-Time graph



This graph shows the location of objects at different times

The horizontal axis represents time

The vertical axis represents displacement

The slope represents velocity

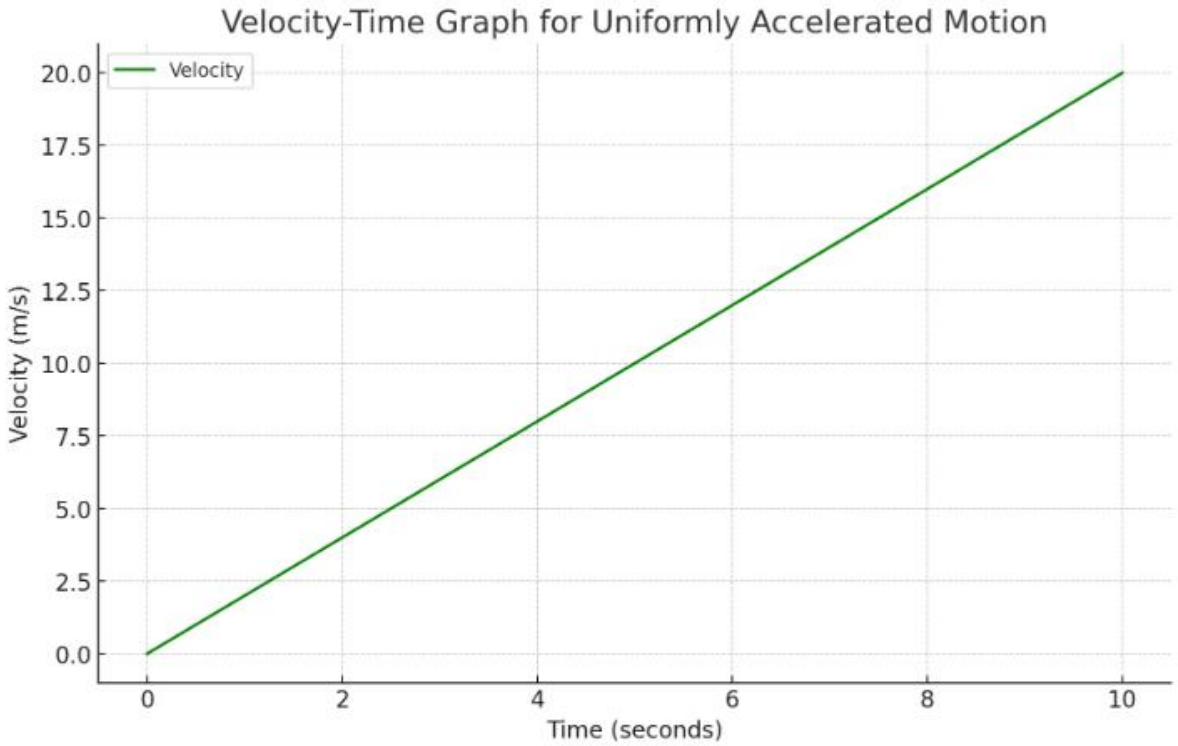
$$v = \frac{\Delta x}{\Delta t}$$

When the graph is a straight line, it indicates that the velocity is constant.

1.3 Representing Motion

The Graph:

2) Velocity-Time graph



The graph shows the velocity of an object at different times

The horizontal axis represents time

The vertical axis represents velocity

The slope represents acceleration

$$\vec{a} = \frac{\Delta v}{\Delta t}$$

The area represents displacement.

$$\Delta x = \vec{v} \cdot t$$

When the graph is a straight line, it indicates that the acceleration is constant.

1.3 Representing Motion

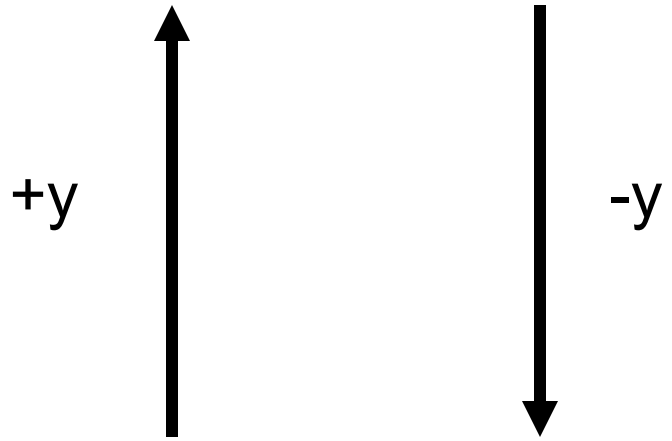
Freely Falling Objects:

Neglecting air resistance, all objects dropped near the Earth's surface fall toward the Earth with the same constant acceleration under the influence of the Earth's gravity.

A freely falling object is any object moving freely under the influence of gravity alone, regardless of its initial motion.

Free-fall acceleration (g) = 9.80 m/s^2

For making quick estimates, use g = 10 m/s^2



$$a_y = -g = -9.80 \text{ m/s}^2$$

1.3 Representing Motion

Freely Falling Objects:

Example 4:

A ball is tossed straight up at 25 m/s. Estimate its velocity at 1 s intervals.

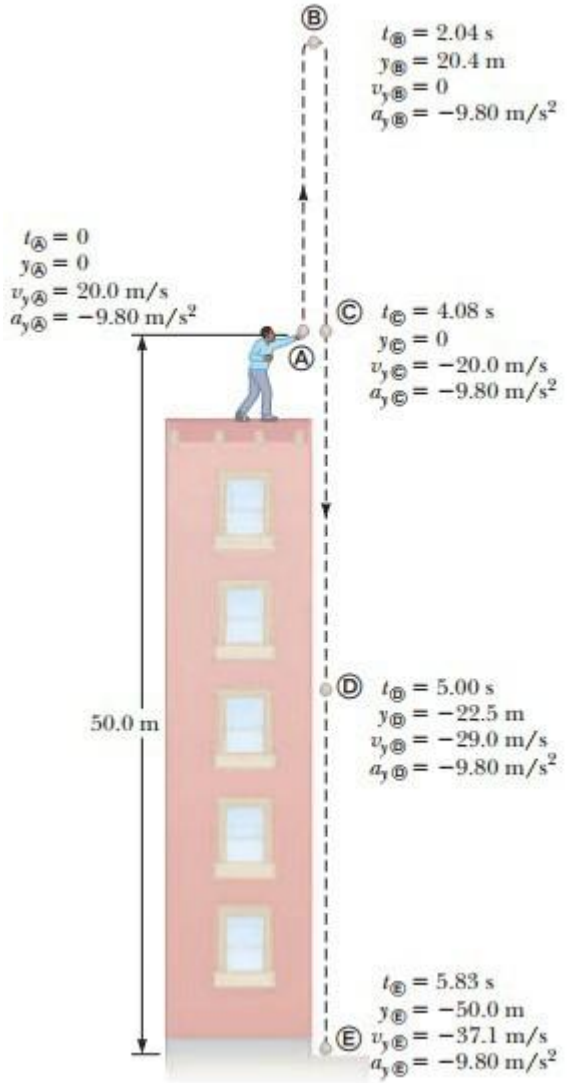
Time (s)	Velocity (m/s)
0	+25
1	
2	
3	
4	
5	

1.3 Representing Motion

Freely Falling Objects:

Example 5:

A stone thrown from the top of a building is given an initial velocity of 20.0 m/s straight upward. The stone is launched 50.0 m above the ground, and the stone just misses the edge of the roof on its way down as shown in Fig.



1) Use $t_A = 0$, as the time the stone leaves the thrower's hand at position A, determine the time at which the stone reaches its maximum height.

2) Find the maximum height of the stone.

3) Determine the velocity of the stone when it returns to the height from which it was thrown.

4) Find the velocity and position of the stone at t = 5.00s

SOLUTION

(a) Find the time when the ball reaches its maximum height.

Write the velocity and position kinematic equations:

$$v = at + v_0$$
$$\Delta y = y - y_0 = v_0 t + \tfrac{1}{2}at^2$$

Substitute $a = -9.80\ \text{m/s}^2$, $v_0 = 20.0\ \text{m/s}$, and $y_0 = 0$ into the preceding two equations:

$$\textbf{(1)}\quad v = (-9.80\ \text{m/s}^2)t + 20.0\ \text{m/s}$$
$$\textbf{(2)}\quad y = (20.0\ \text{m/s})t - (4.90\ \text{m/s}^2)t^2$$

Substitute $v = 0$, the velocity at maximum height, into Equation (1) and solve for time:

$$0 = (-9.80\ \text{m/s}^2)t + 20.0\ \text{m/s}$$
$$t = \frac{-20.0\ \text{m/s}}{-9.80\ \text{m/s}^2} = \boxed{2.04\ \text{s}}$$

(b) Determine the ball's maximum height.

Substitute the time $t = 2.04$ s into Equation (2):

$$y_{\text{max}} = (20.0\ \text{m/s})(2.04\ \text{s}) - (4.90\ \text{m/s}^2)(2.04\ \text{s})^2 = \boxed{20.4\ \text{m}}$$

(c) Find the time the ball takes to return to its initial position, and find the velocity of the ball at that time.

Set $y = 0$ in Equation (2) and solve for t:

$$0 = (20.0\ \text{m/s})t - (4.90\ \text{m/s}^2)t^2$$
$$= t(20.0\ \text{m/s} - (4.90\ \text{m/s}^2)t)$$
$$t = \boxed{4.08\ \text{s}}$$

Substitute the time into Equation (1) to get the velocity:

$$v = 20.0\ \text{m/s} + (-9.80\ \text{m/s}^2)(4.08\ \text{s}) = -20.0\ \text{m/s}$$

(d) Find the time required for the ball to reach the ground.

In Equation (2), set $y = -50.0$ m:

$$-50.0\ \text{m} = (20.0\ \text{m/s})t - (4.90\ \text{m/s}^2)t^2$$

Apply the quadratic formula and take the positive root:

$$t = 5.83\ \text{s}$$

(e) Find the velocity and position of the ball at $t = 5.00$ s.

Substitute values into Equations (1) and (2):

$$v = (-9.80\ \text{m/s}^2)(5.00\ \text{s}) + 20.0\ \text{m/s} = -29.0\ \text{m/s}$$

$$y = (20.0\ \text{m/s})(5.00\ \text{s}) - (4.90\ \text{m/s}^2)(5.00\ \text{s})^2 = -22.5\ \text{m}$$

Find the x-component of velocity:

$$0 = v_x - 5.00 \text{ km/h} \rightarrow v_x = 5.00 \text{ km/h}$$

Find the y-component of velocity:

$$10.0 \text{ km/h} = v_y - 0 \rightarrow v_y = 10.0 \text{ km/h}$$

Find the magnitude of $\vec{\mathbf{v}}_{\text{BE}}$:

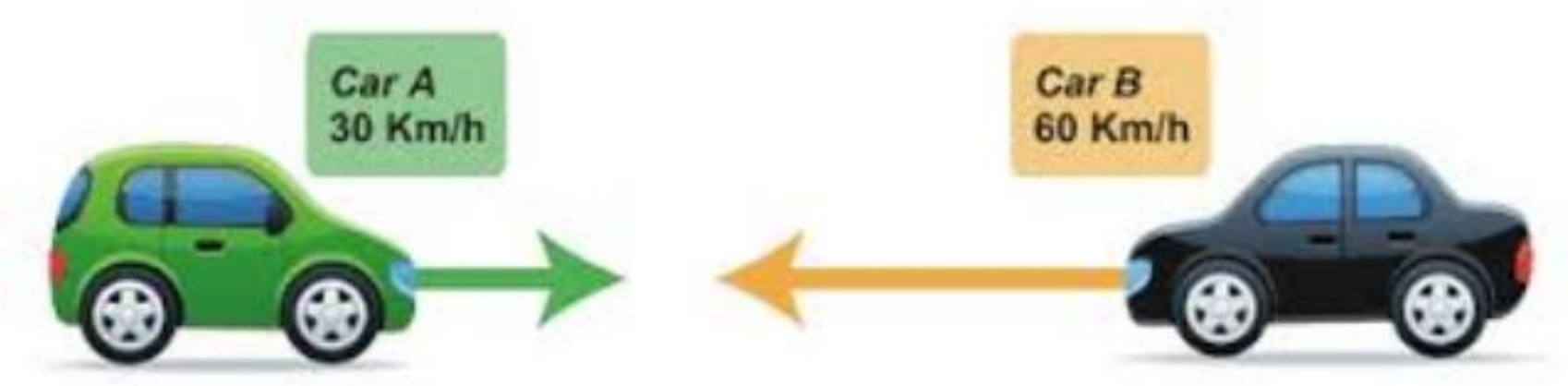
$$v_{\text{BE}} = \sqrt{v_x^2 + v_y^2}$$
$$= \sqrt{(5.00 \text{ km/h})^2 + (10.0 \text{ km/h})^2} = \boxed{11.2 \text{ km/h}}$$

Find the direction of $\vec{\mathbf{v}}_{\text{BE}}$:

$$\theta = \tan^{-1}\left(\frac{v_x}{v_y}\right) = \tan^{-1}\left(\frac{5.00 \text{ m/s}}{10.0 \text{ m/s}}\right) = \boxed{26.6^\circ}$$

1.4 Reference Frames and Relative Motion

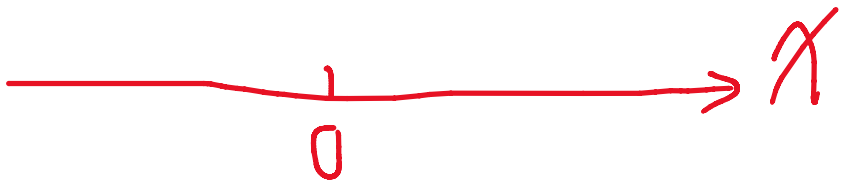
In order to describe physical events that occur in space and time such as the motion of bodies, we introduced a coordinate system. Its spatial and temporal coordinates can now specify a space-time event. In particular, the position of a moving body can be described by space-time events specified by its space-time coordinates. You can place an observer at the origin of coordinate system. The coordinate system with your observer acts as a reference frame for describing the position, velocity, and acceleration of bodies. The position vector of the body depends on the choice of origin (location of your observer) but the displacement, velocity, and acceleration vectors are independent of the location of the observer.



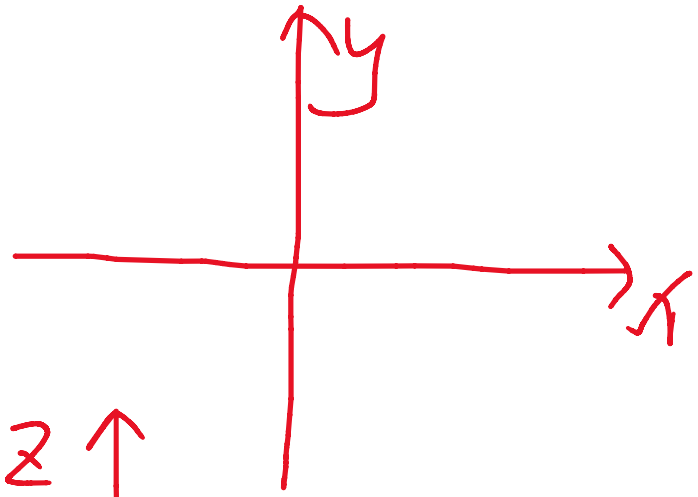
1.4 Reference Frames and Relative Motion

Reference Frame:

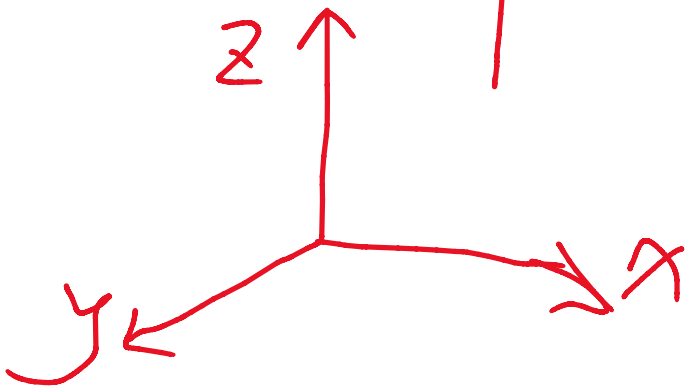
- 1D



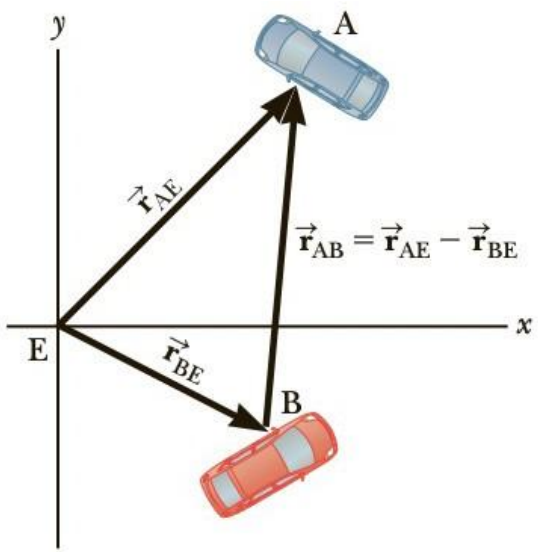
- 2D



- 3D



- Relative velocity is all about relating the measurements of two different observers, one moving with respect to the other.
- The measured velocity of an object depends on the velocity of the observer with respect to the object.
- On highways, for example, cars moving in the same direction are often moving at high-speed relative to Earth, but relative to each other they hardly move at all.
- To an observer at rest at the side of the road, a car might be traveling at 80km/h, but to an observer in a truck traveling in the same direction at 70 km/h, the car would appear to be traveling only 10 km/h.



$\vec{\mathbf{r}}_{AE}$ = the position of Car A as measured by E (in a coordinate system fixed with respect to Earth).

$\vec{\mathbf{r}}_{BE}$ = the position of Car B as measured by E.

$\vec{\mathbf{r}}_{AB}$ = the position of Car A as measured by an observer in Car B.

- According to the preceding notation, the first letter tells us what the vector is pointing at and the second letter tells us where the position vector starts.
- The position vectors of Car A and Car B relative to E, $\vec{r}_{AE}$ and $\vec{r}_{BE}$ are given in the figure. How do we find $\vec{r}_{AB}$ the position of Car A as measured by an observer in Car B? We simply draw an arrow pointing from Car B to Car A, which can be obtained by subtracting $\vec{r}_{BE}$ from $\vec{r}_{AE}$:

$$\vec{\mathbf{r}}_{AB} = \vec{\mathbf{r}}_{AE} - \vec{\mathbf{r}}_{BE} \qquad [3.10]$$

Now, the rate of change of these quantities with time gives us the relationship between the associated velocities:

$$\vec{\mathbf{v}}_{AB} = \vec{\mathbf{v}}_{AE} - \vec{\mathbf{v}}_{BE} \qquad [3.11]$$

Example 4.8 A Boat Crossing a River

A boat crossing a wide river moves with a speed of 10.0 km/h relative to the water. The water in the river has a uniform speed of 5.00 km/h due east relative to the Earth.

(A) If the boat heads due north, determine the velocity of the boat relative to an observer standing on either bank.

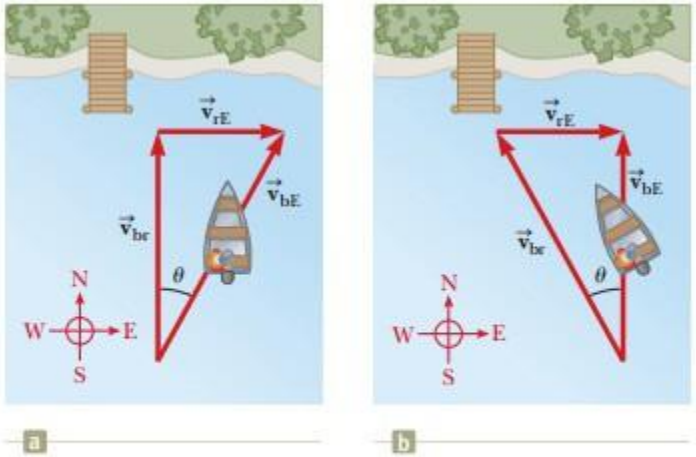


Figure 4.21 (Example 4.8) (a) A boat aims directly across a river and ends up downstream. (b) To move directly across the river, the boat must aim upstream.

(B) If the boat travels with the same speed of 10.0 km/h relative to the river and is to travel due north as shown in Figure 4.21b, what should its heading be?

SOLUTION

Arrange the three quantities into the proper relative velocity equation:

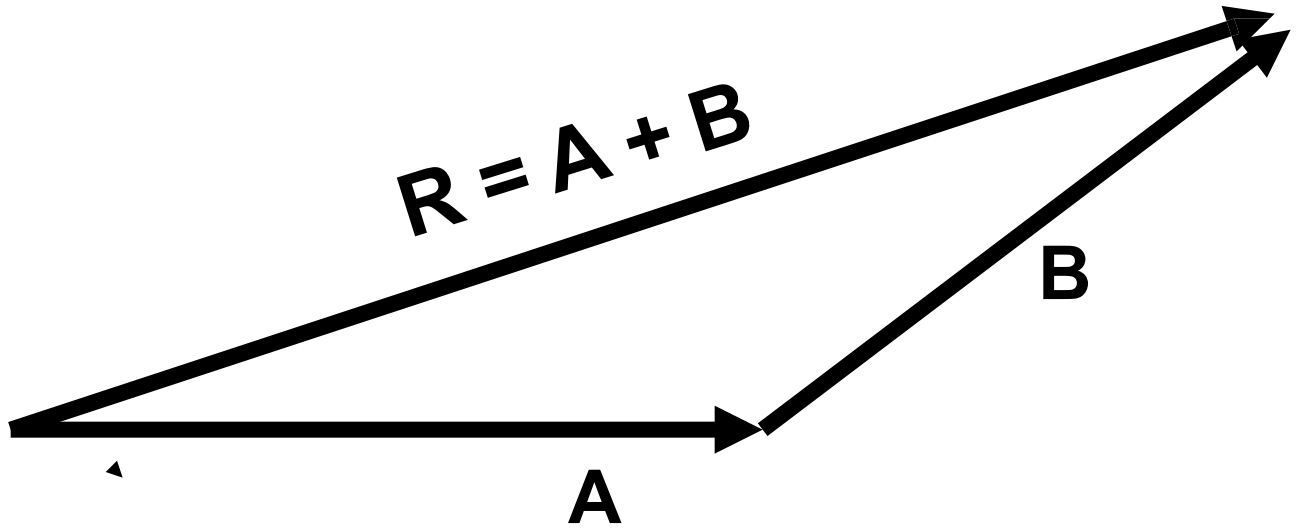
Write the velocity vectors in terms of their components. For convenience, these are organized in the following table:

$$\vec{\mathbf{v}}_{BR} = \vec{\mathbf{v}}_{BE} - \vec{\mathbf{v}}_{RE}$$

Vector	x-Component (km/h)	y-Component (km/h)
$\vec{\mathbf{v}}_{BR}$	0	10.0
$\vec{\mathbf{v}}_{BE}$	v_x	v_y
$\vec{\mathbf{v}}_{RE}$	5.00	0

1.5 Vectors and Motion in Two Dimensions

Adding Vectors

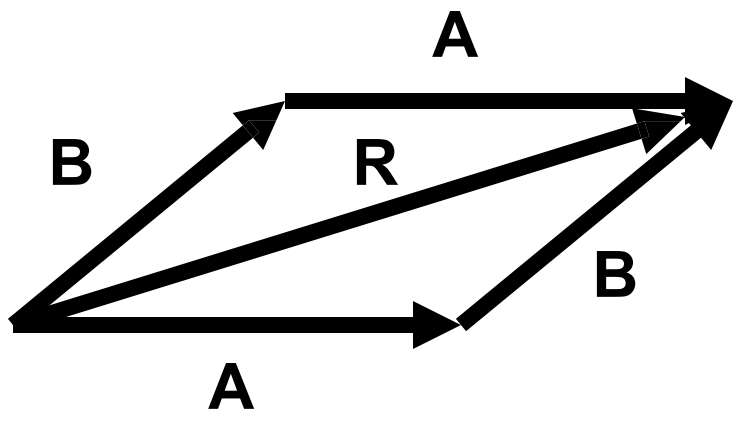


Resultant (R) is drawn from the tail of the first vector to the tip of the last vector

Commutative Law of Addition

When two vectors are added, the sum is independent of the order of the addition.

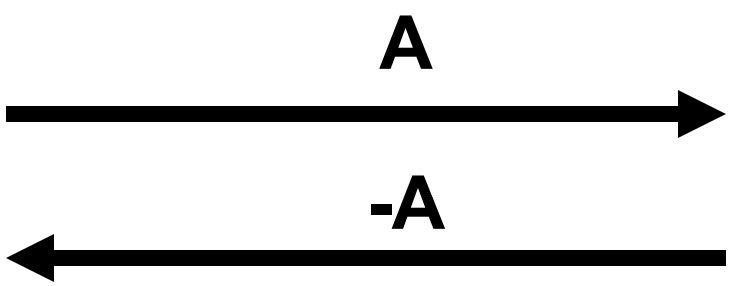
$$A + B = B + A$$



Negative of a Vector

The vector that when added to A gives zero for the vector sum.

$$A + (-A) = 0$$

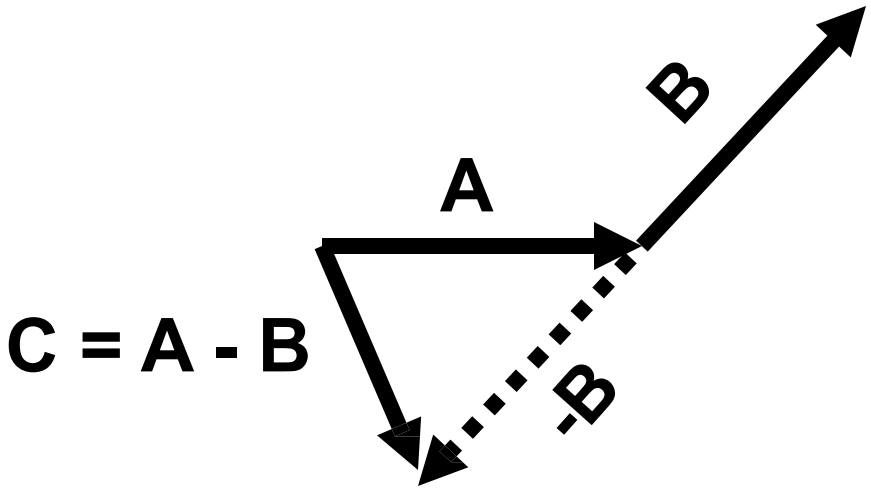


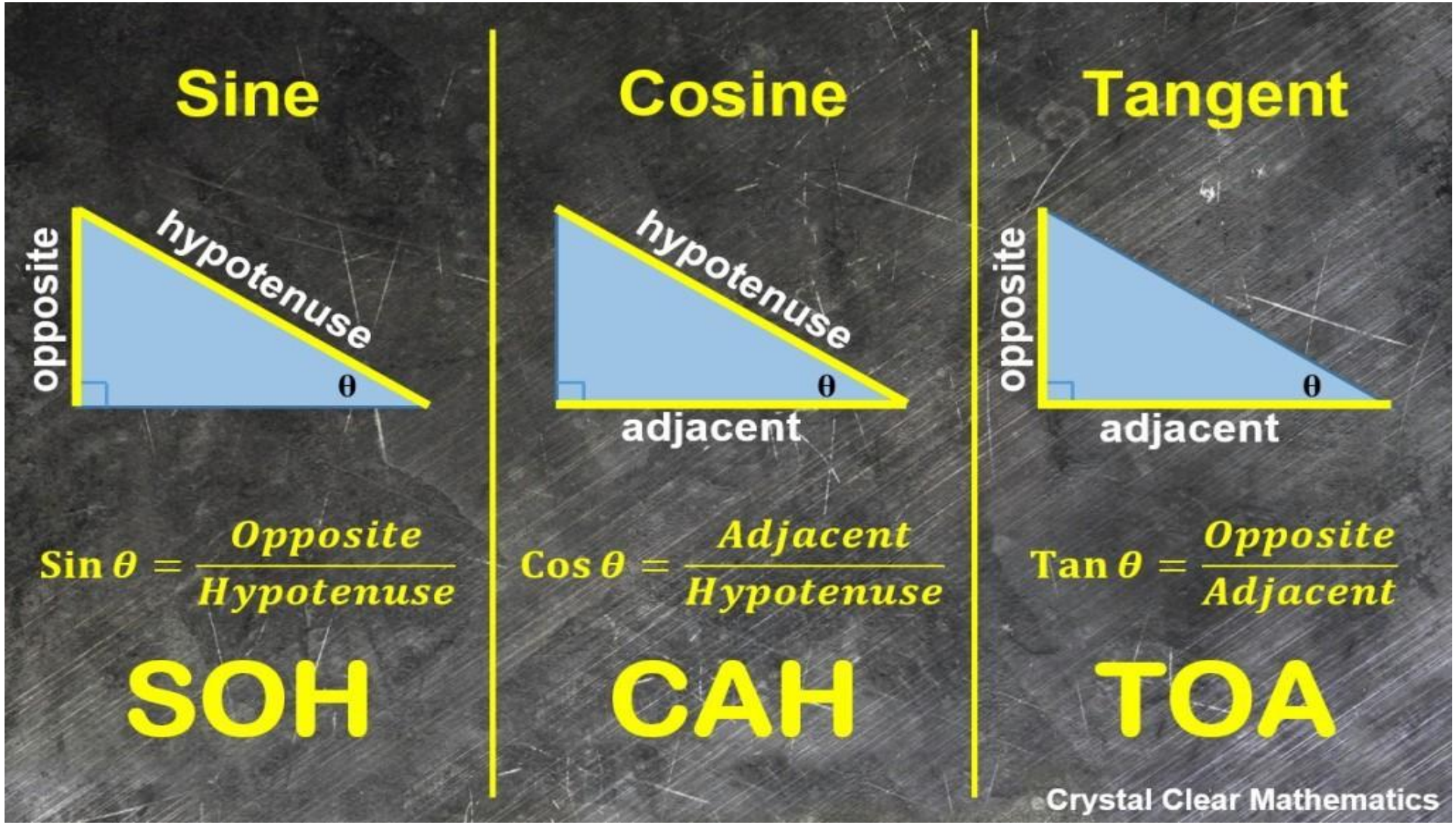
A and –A have the same magnitude but point in opposite directions

Subtracting Vectors

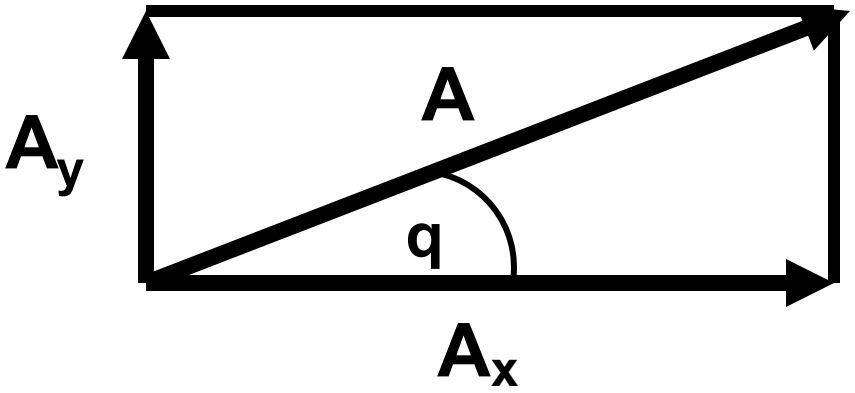
We define the operation A – B as vector –B added to vector A.

$$A - B = A + (-B)$$




Sine
opposite
hypotenuse
θ
$\text{Sin}\,\theta = \frac{Opposite}{Hypotenuse}$
SOH
Cosine
hypotenuse
θ
adjacent
$\text{Cos}\,\theta = \frac{Adjacent}{Hypotenuse}$
CAH
Tangent
opposite
θ
adjacent
$\text{Tan}\,\theta = \frac{Opposite}{Adjacent}$
TOA
Crystal Clear Mathematics

Components of a Vector and Unit Vectors



$$A = Ax + Ay$$

$$\boxed{A_x = A \cos q}$$

$$\boxed{A_y = A \sin q}$$

Component Form:

$$A_x = A\cos\theta$$

$$A_y = A\sin\theta$$

Magnitude & Direction Form:

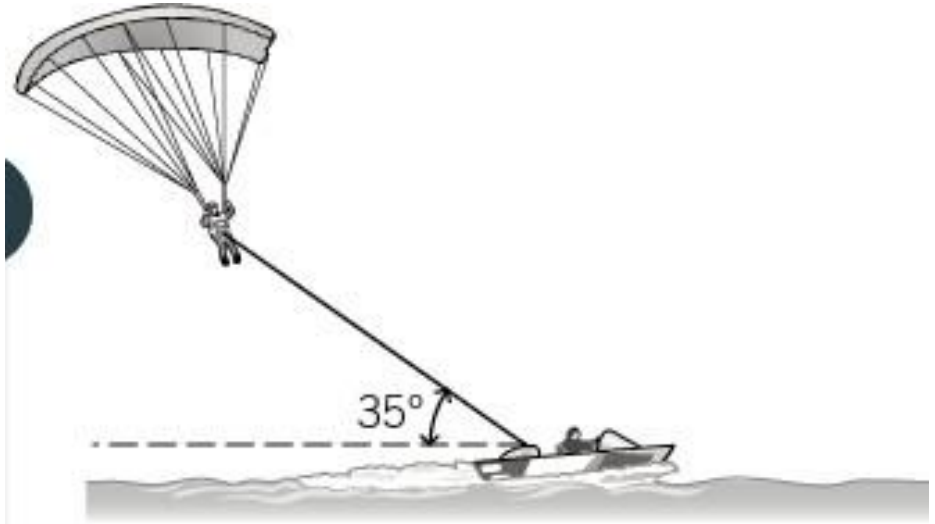
$$A = \sqrt{A_x^2 + A_y^2}$$

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$$

But Be careful! Cos and sin might be reversed. You always have to look at your right triangles carefully.

Worked example

A paraglider is pulled along at constant height at steady speed by a cable attached to a speedboat as shown in Figure 5. The cable pulls on the paraglider with a force of 500 N at an angle of 35° to the horizontal. Calculate the horizontal and vertical components of this force.



Solution

Because the force on the paraglider is at an angle of 35° below the horizontal, the horizontal and vertical components of this force are:

- $500 \cos 35° = 410\,\text{N}$ horizontally to the right
- $500 \sin 35° = 287\,\text{N}$ vertically downwards.

Projectile motion

A projectile is an object projected into the air and acted upon only by the forces of gravity and air resistance.

Initially, we assume air resistance is negligible. The key principles when air resistance is negligible areas follows:

- The acceleration of the object is always equal to g and is always downwards because the force of gravity acts downwards. The acceleration therefore only affects the vertical motion of the object.
- The horizontal velocity of the object is constant because the acceleration of the object does not have a horizontal component.

Projectile motion

- The motions in the horizontal and vertical directions are independent of each other.

Vertical projection

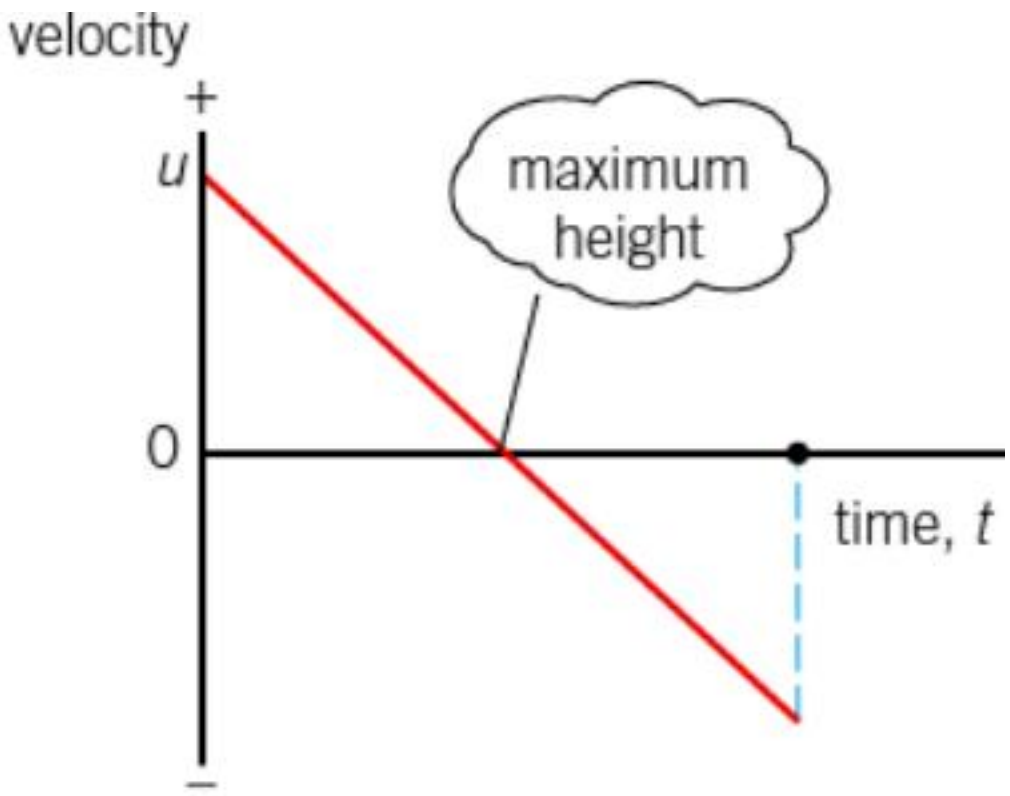
Such an object moves vertically as it has no horizontal motion. Its acceleration is 9.8 ms^{-2} downwards. Using the direction code (+) for upwards and direction code (–) for downwards, its displacement, *y*, and velocity, *v* after time *t* are given by

$$v = u - gt$$

$$y = ut - \frac{1}{2}gt^2$$

Projectile motion

Vertical projection



▲ **Figure 1** *Upward projection*

Projectile motion

Horizontal projection

- If its initial velocity is U, then at time t after projection:
- The horizontal component of its displacement,

$$x = Ut$$

Because no acceleration in the horizontal direction

- The vertical component of its displacement,

$$y = \frac{1}{2}gt^2$$

Because it has no vertical component of its initial velocity

Vertical and horizontal at the same time

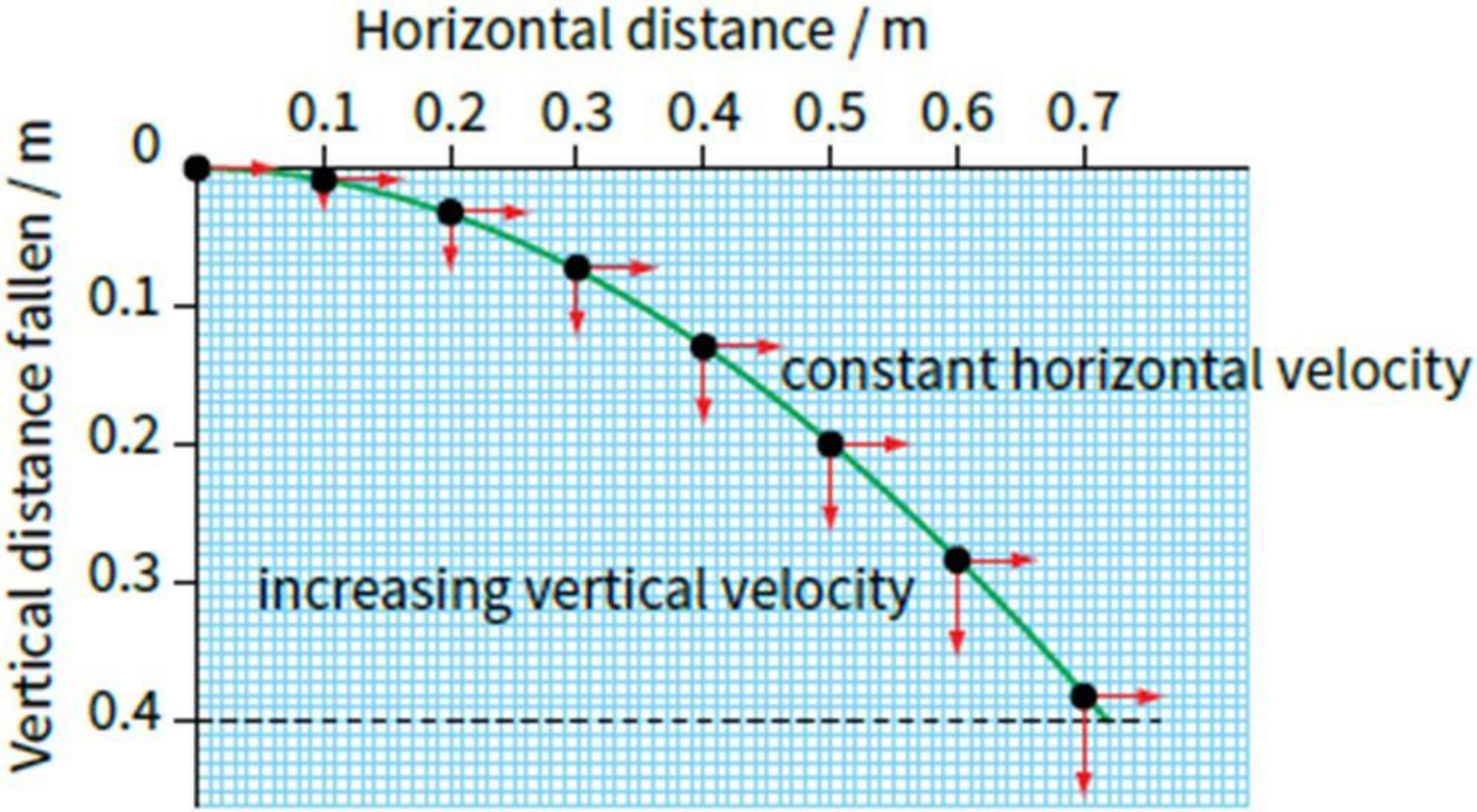


Figure 2.31 This sketch shows the path of the ball projected horizontally. The arrows represent the horizontal and vertical components of its velocity.

Horizontal projection

- Its velocity has
  1. a horizontal component $v_x = U$ and
  2. Vertical component $v_y = -gt$ (from $v = u + at$)

- Note that at time t, the speed $=(v_x^2 + v_y^2)^{\frac{1}{2}}$

Worked example

$g = 9.8\,\mathrm{m\,s^{-2}}$

An object is projected horizontally at a speed of $15\,\mathrm{m\,s^{-1}}$ from the top of a tower of height 35.0 m. Calculate:

a how long it takes to fall to the ground
b how far it travels horizontally
c its speed just before it hits the ground.

Example 4.1 Motion in a Plane AM

A particle moves in the xy plane, starting from the origin at $t = 0$ with an initial velocity having an x component of 20 m/s and a y component of −15 m/s. The particle experiences an acceleration in the x direction, given by $a_x = 4.0\ \text{m/s}^2$.

(A) Determine the total velocity vector at any time.

(B) Calculate the velocity and speed of the particle at $t = 5.0$ s and the angle the velocity vector makes with the x axis.

(C) Determine the x and y coordinates of the particle at any time t and its position vector at this time.

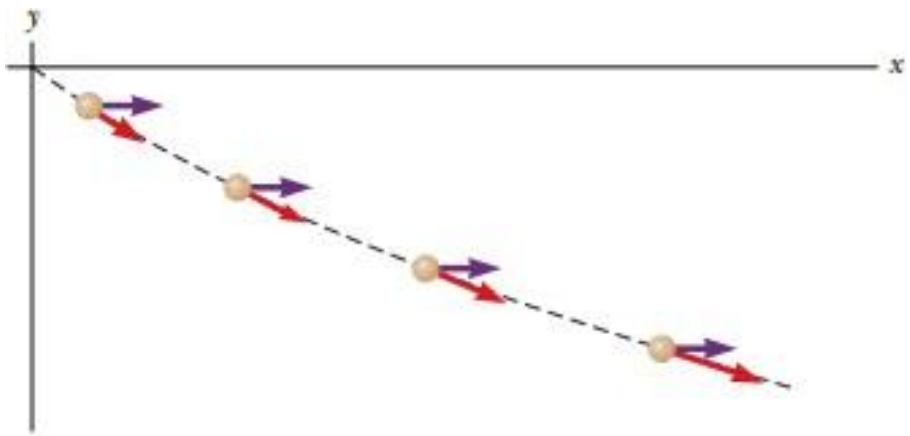


Figure 4.6 (Example 4.1) Motion diagram for the particle.

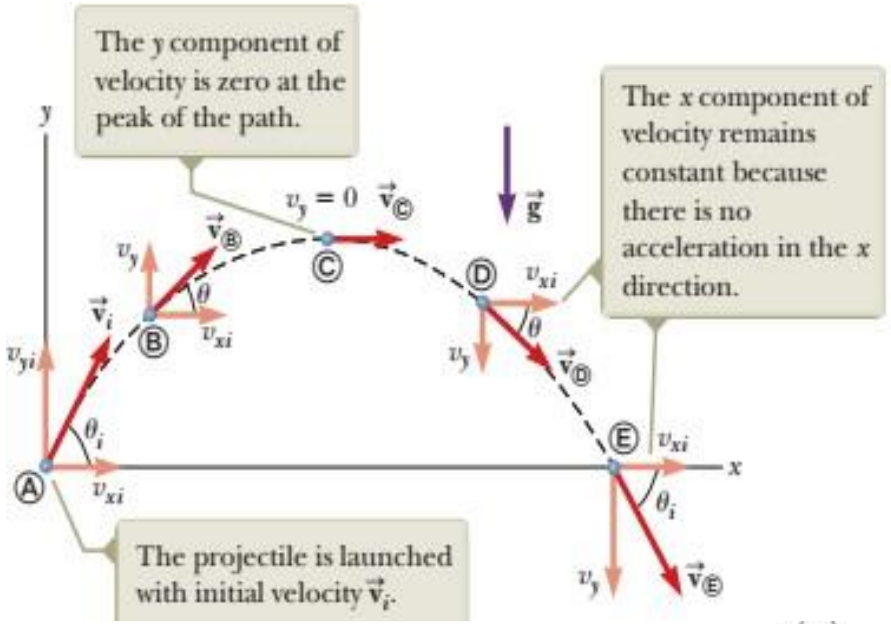


Figure 4.7 The parabolic path of a projectile that leaves the origin with a velocity $\vec{v}_i$. The velocity vector $\vec{v}$ changes with time in both magnitude and direction. This change is the result of acceleration $\vec{a} = \vec{g}$ in the negative y direction.

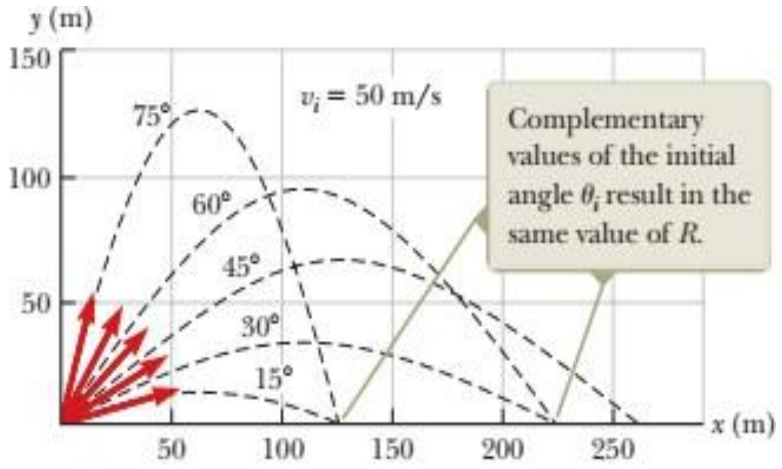
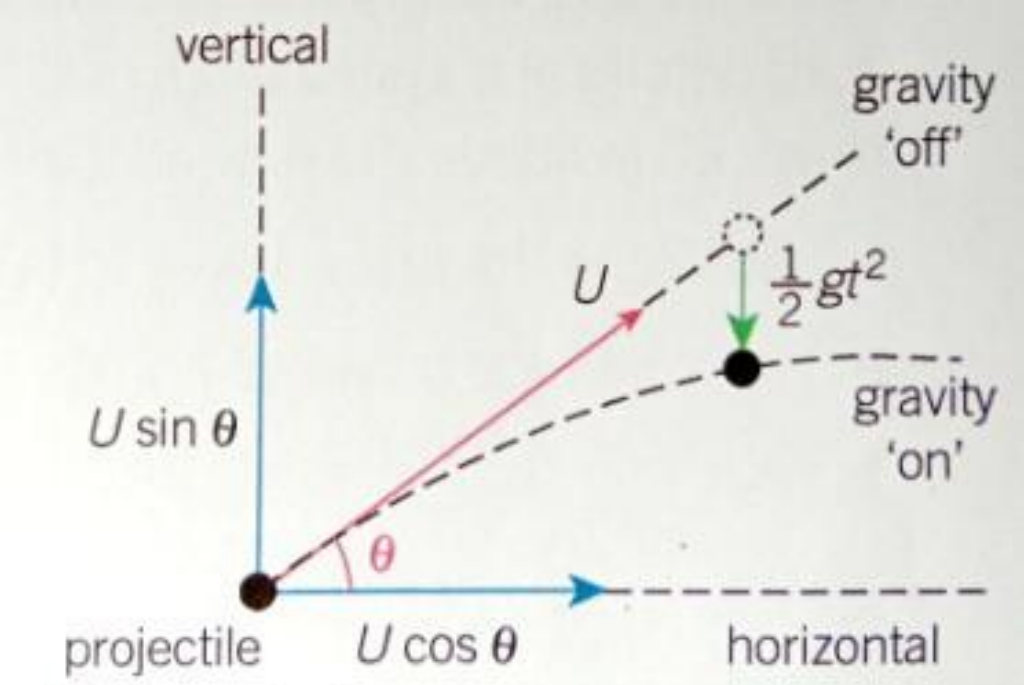


Figure 4.10 A projectile launched over a flat surface from the origin with an initial speed of 50 m/s at various angles of projection.

Imagination at work

- Suppose gravity could be switched off, with air resistance negligible.
- An object projected into air would follow a straight line with no change in its velocity (figure 4)



▲ **Figure 4** *Imagination at work*

Imagination at work

- Suppose it had been launched at a speed U in a direction at angle θ above the horizontal, its initial velocity would therefore have a horizontal component U cos θ and a vertical component U sin θ.
- Without gravity its velocity would be unchanged, so its displacement at time t after projection would have
  - A horizontal component $x = Ut\cos\theta$
  - A vertical component $y = Ut\sin\theta$

Imagination at work

- Let's now consider the difference made to its displacement and velocity if we bring back gravity. At a time *t* after the launch,
  - Its vertical position with gravity on would be $\frac{1}{2}gt^2$ lower

$$y = Ut\sin\theta - \frac{1}{2}gt^2$$

  - Its vertical component of velocity would be changed by $-gt$ from its initial value $U\ sin\ \theta$.

$$v_y = U\sin\theta - gt$$

Maximum Height of a Projectile

$$v_y = v_{0y} - gt$$

$$0 = v_0 \sin q - gt \quad \text{(at peak)}$$

$$t = \frac{v_0 \sin q}{g} \quad \text{(at peak)}$$

$$Dy = v_{0y}\, t - \tfrac{1}{2} g t^2$$

$$h = (v_0 \sin q)\frac{v_0 \sin q}{g} - \tfrac{1}{2} g \left(\frac{v_0 \sin q}{g}\right)^2$$

$$\boxed{h = \frac{v_0{}^2 \sin^2 q}{2g}}$$

Horizontal Range of a Projectile

$$Dx = R = U_x\ t$$

$$R = U \cos q\ 2t \text{ (twice peak time)}$$

$$t = \frac{U \sin q}{g} \quad \text{(at peak)}$$

$$R = U \cos q\ \frac{2U \sin q}{g}$$

$$\sin 2q = 2 \sin q \cos q \text{ (trig identity)}$$

$$\boxed{R = \frac{U^2 \sin 2\theta}{g}}$$

Example 4.2 The Long Jump

A long jumper (Fig. 4.11) leaves the ground at an angle of 20.0° above the horizontal and at a speed of 11.0 m/s.

(A) How far does he jump in the horizontal direction?

(B) What is the maximum height reached?

The long jump solution

(a) Find the horizontal distance he jumps.

First find the time for the jump, which is twice t_{max}:

Substitute this result into the equation for the x-displacement:

$$t = 2t_{max} = 2(0.384\ \text{s}) = 0.768\ \text{s}$$

$$\textbf{(2)} \quad \Delta x = (v_0 \cos\theta_0)t$$
$$= (11.0\ \text{m/s})(\cos 20.0°)(0.768\ \text{s}) = \boxed{7.94\ \text{m}}$$

The long jump solution

(b) Find the time t_{max} taken to reach maximum height.

$$v_y = v_0 \sin\theta_0 - g t_{max} = 0$$

$$\textbf{(1)} \quad t_{max} = \frac{v_0 \sin\theta_0}{g}$$

$$= \frac{(11.0\ \text{m/s})(\sin 20.0°)}{9.80\ \text{m/s}^2} = \boxed{0.384\ \text{s}}$$

Find the maximum height he reaches.

Substitute the time t_{max} into the equation for the y-displacement,

$$t = 2t_{max} = 2(0.384\ \text{s}) = 0.768\ \text{s}$$

$$\textbf{(2)} \quad \Delta x = (v_0 \cos\theta_0)t$$

$$= (11.0\ \text{m/s})(\cos 20.0°)(0.768\ \text{s}) = \boxed{7.94\ \text{m}}$$

Example 4.4 That's Quite an Arm! AM

A stone is thrown from the top of a building upward at an angle of 30.0° to the horizontal with an initial speed of 20.0 m/s as shown in Figure 4.13. The height from which the stone is thrown is 45.0 m above the ground.

(A) How long does it take the stone to reach the ground?

(B) What is the speed of the stone just before it strikes the ground?

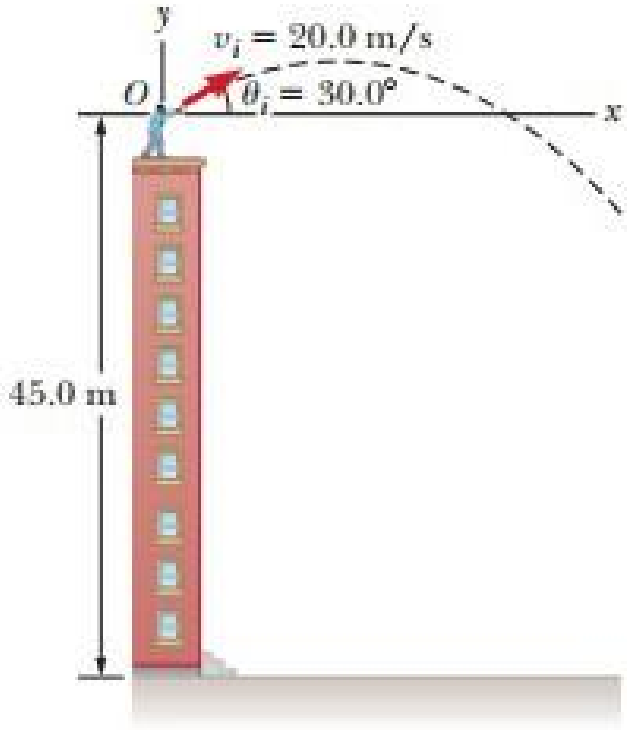


Figure 4.13 (Example 4.4) A stone is thrown from the top of a building.

Example 4.5 The End of the Ski Jump AM

A ski jumper leaves the ski track moving in the horizontal direction with a speed of 25.0 m/s as shown in Figure 4.14. The landing incline below her falls off with a slope of 35.0°. Where does she land on the incline?

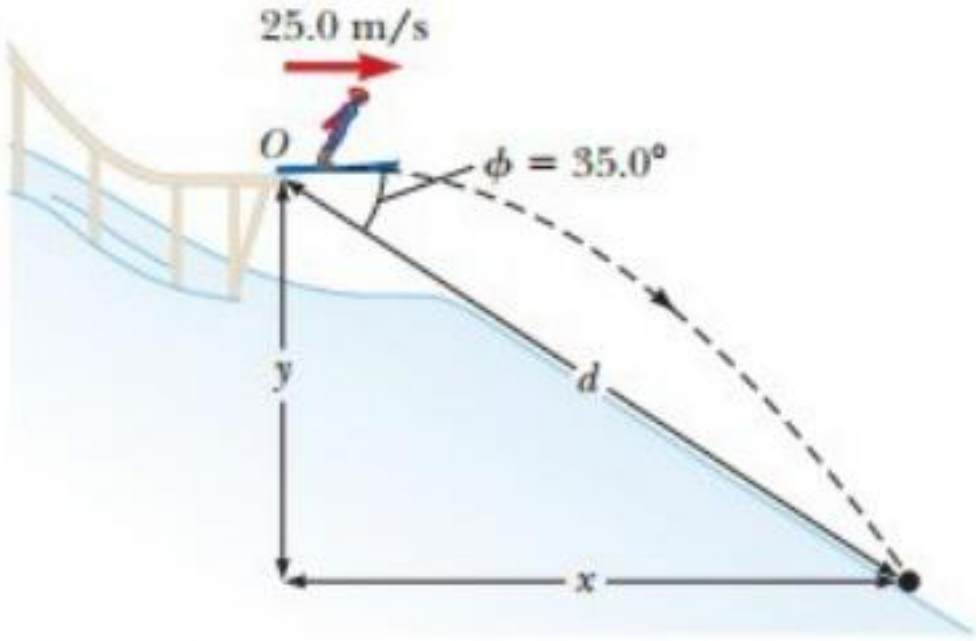
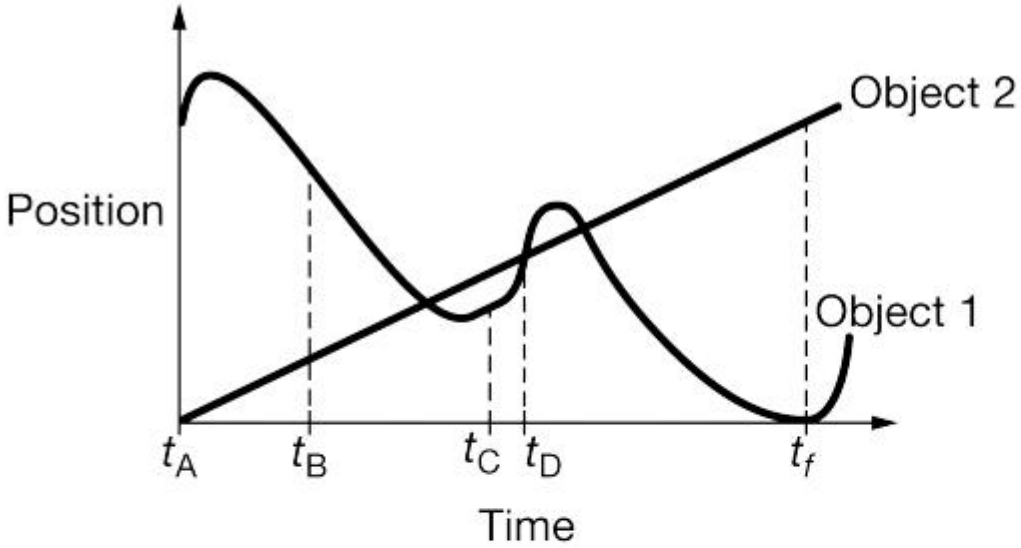


Figure 4.14 (Example 4.5) A ski jumper leaves the track moving in a horizontal direction.

Example Question of AP Classroom



The position as a function of time for two objects moving along a straight line is shown in the graph.

1. Which statement is true about the distances the two object have traveled at time t_f?

(A) Object 1 has traveled a greater distance.

(B) Object 2 has traveled a greater distance.

(C) Both objects have traveled the same distance.

(D) The total distance traveled by each object cannot be compared using the graph.

Example Question of AP Classroom

A cart is moving along a straight track and has velocity $+0.5\ \mathrm{m/s}$ and positive acceleration at time $1\ \mathrm{s}$. At time $3\ \mathrm{s}$, the cart reaches a maximum speed. At time $4\ \mathrm{s}$, the cart again has velocity $+0.5\ \mathrm{m/s}$, but with negative acceleration. A student claims that the average acceleration of the cart between time $1\ \mathrm{s}$ and $4\ \mathrm{s}$ time is zero. Is the student's claim correct? Why or why not?

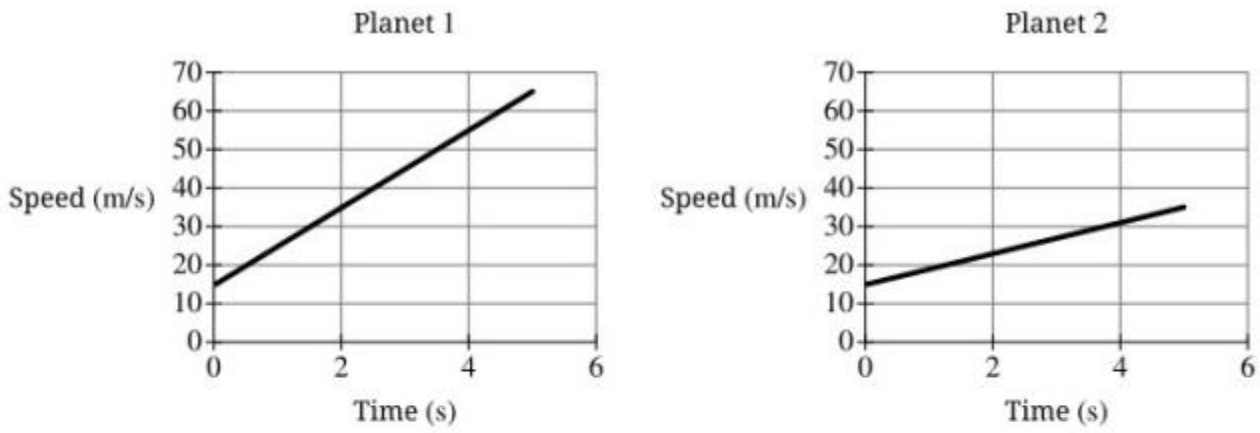
(A) Yes, because the final velocity of the cart is equal to the initial velocity.

(B) Yes, because the cart returns to its original position.

(C) No, because the values of the acceleration at time $1\ \mathrm{s}$ and time $4\ \mathrm{s}$ are unknown.

(D) No, because the cart spends more time with a positive acceleration than with a negative acceleration.

Example Question of AP Classroom



The graphs represent the speed of a ball thrown downward as a function of time near the surface of two planets. Which of the following correctly relates the acceleration a_1 of the ball near the surface of Planet 1 and the acceleration a_2 of the ball near the surface of Planet 2?

(A) $a_1 > a_2 > 0$

(B) $a_2 > a_1 > 0$

(C) $(a_1 = a_2) > 0$

(D) $a_1 = a_2 = 0$